

RESEARCH ARTICLE

Functional Ecology



Distinguishing between unreliability and dishonesty: A comparative study of aggressive communication in crayfish

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Abstract

1. A major challenge in the study of animal communication is distinguishing whether signals convey honest or dishonest information. Biologists infer the honesty of a signal from its correlation with the information being signalled (e.g. fecundity or fighting ability)—the better the correlation, the more honest the signal.
2. However, this view of signalling potentially conflates unreliable indicators with dishonest signals. Just because a trait conveys unreliable information does not mean that the structure serves as a dishonest signal; developmental noise, genetic drift and environmental constraints can reduce the covariation between a putative signal and an organism's quality. Moreover, a trait must influence the behaviour of other organisms to qualify as a signal.
3. We studied how a putative signal, claw size, affects physical performance and social dominance in three species of stream-dwelling crayfish, which fight routinely over resources. For comparison, we measured the relationship between claw size and claw strength in three species of burrowing crayfish, which do not fight.
4. In all species, crayfish with larger claws were not necessarily stronger, indicating that claw size poorly indicates claw strength. However, claw size is unlikely to function as a dishonest signal of fighting ability in these species for two reasons. First, claw size unreliably indicates claw strength in both the stream-dwelling species and the burrowing species. Second, relative claw size poorly predicted whether stream-dwelling crayfish would escalate aggression, questioning whether claw size serves as a signal. Instead, stream-dwelling crayfish escalated aggression based on their relative body size.
5. Our results highlight the importance of observing behaviour when determining the honesty of a signal. As such, future studies must distinguish between phenotypes that are unreliable indicators (or reliable indicators) from phenotypes that have been selected for dishonest (or honest) signalling.

KEYWORDS

aggression, communication, deception, dishonest signals, signal reliability

1 | INTRODUCTION

A major goal of animal communication research is to determine whether signals convey honest or dishonest information. An honest signal can be defined as a reliable relationship between an organism's phenotype (e.g. a structure or behaviour) and unobserved quality of the signaller (e.g. competitive ability; Hughes, 2000). By contrast, a dishonest signal occurs when selection erodes this relationship such that signallers deceive receivers. Traditionally, biologists believed that dishonest communication should be rare or absent among animals (Dawkins & Krebs, 1978; Maynard Smith & Harper, 1983; Searcy & Nowicki, 2005). This view emerged from theoretical models of how communication evolves when senders and receivers have conflicting goals. According to these models, dishonest signals decline in frequency, because receivers benefit from ignoring signals that convey unreliable information (Adams & Mesterton-Gibbons, 1995; Enquist, 1985; Grafen, 1990; Számadó, 2000). However, the same models established that dishonest signals can persist under special conditions: (a) when constraints prevent one from easily detecting unreliable information, and (b) when ignoring a reliable signal carries a great cost (Adams & Mesterton-Gibbons, 1995). These conditions can explain why a recently moulted mantis shrimp will aggressively defend its burrow despite being unable to harm intruders; in this case, the mantis shrimp's condition cannot be confirmed by an intruder without risking injury (Adams & Caldwell, 1990). In fact, the majority of signalling in a population could be dishonest if signallers benefit greatly from this behaviour (Botero et al., 2010; Searcy & Nowicki, 2005; Szalai & Számadó, 2009; Számadó, 2008; Wagner, 1992).

Biologists must accomplish two goals to demonstrate dishonest signalling: (a) quantify the covariation between a putative dishonest

signal and the quality being signalled, and (b) establish that receivers respond to the putative signal in ways that benefit a dishonest signaller (Graham et al., 2020; Hughes, 2000). The covariation between putative signals and quality has been used to infer dishonesty of diverse forms of communication, including begging calls in chicks (Caro et al., 2016) and territorial marks in dogs (McGuire et al., 2018). However, in many cases, evidence supports only a potential for dishonest signalling, because the behavioural responses of receivers and their implications for signallers remain unknown. Without this behavioural evidence, one can conclude that a trait is an unreliable indicator but cannot conclude that it functions as a dishonest signal. We define a trait to be an indicator based on its correlation to some observable and measurable quality. Studies of scarab beetles highlight the importance of distinguishing between an indicator and a signal (Figure 1). Only scarab beetles in the best nutritional condition can develop enlarged horns, and these horns are presumably used as signals of the wielder's quality (McCullough et al., 2016; Warren et al., 2013). However, no evidence supports the conclusion that horns function as signal. Scarab beetles are nocturnal and have reduced eyes, making it unlikely that competitors use horn size as a signal. Simply put, all dishonest signals are unreliable indicators, but not all unreliable indicators are dishonest signals (Figure 1a). Likewise, all honest signals are reliable indicators, but not all reliable indicators are honest signals (Figure 1g).

Biologists must study behaviour to confirm that an unreliable indicator evolved for dishonest signalling. By observing how organisms respond to putative signals, we can first confirm that an indicator serves as a signal. Then, by interpreting the response to a signal in the context of its reliability, we can confirm that the signal functions to deceive receivers (Figure 1b) or honestly communicate to receivers (Figure 1h). In some cases, what one presumes to function as

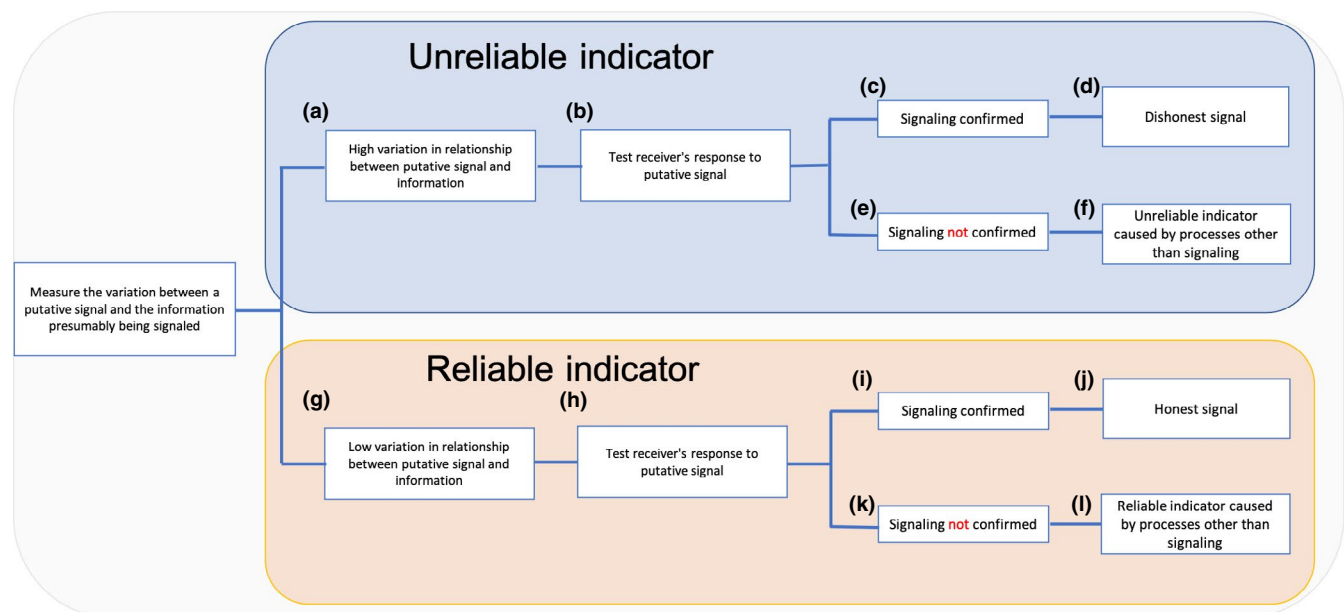


FIGURE 1 Flowchart of the necessary steps to distinguish between (a–f) a dishonest signal and a trait that is an unreliable indicator due to processes other than signalling (i.e. physiological, developmental or environmental constraints) and (g–l) an honest signal and a trait that is a reliable indicator due to processes other than signalling

a dishonest signal could simply be an unreliable indicator, resulting from developmental noise or genetic drift noise (Figure 1d; Botero et al., 2010; Brumm, 2013; Tanner & Bee, 2019; Wagner, 1992). Thus, behavioural studies are critical for distinguishing between whether or not putative signals are simply unreliable indicators or dishonest signals (Figure 1b,h).

Research on crustaceans has provided strong evidence of dishonest signalling (reviewed in Christy & Rittschof, 2011; Wilson & Angilletta, 2015) and highlights the importance of distinguishing between unreliable indicators and dishonest signals. For freshwater crayfish, the size and strength of claws varies substantially among individuals of a species (Graham & Angilletta, 2020; Graham et al., 2020; Malavé et al., 2017; Robinson & Gifford, 2019; Walter et al., 2011; Wilson et al., 2007). In all species of crayfish investigated so far, claw strength varies substantially for a given claw size, fulfilling the first criterion for dishonest singling (Figure 1a). In some species, the largest crayfish can generate pinching forces that vary among individuals by an order of magnitude (Wilson et al., 2007). Still, the second criterion for dishonest signalling, behavioural evidence of receiver response, has mixed support. Behavioural experiments demonstrate that claws function as dishonest signals in only some of these species. For example, in two species of Australian crayfish (*Cherax dispar* and *C. destructor*), weak individuals with large claws can successfully deter aggression from competitors; however, dishonest signallers are likely to lose due to their weak claws when aggression escalates to a fight (Walter et al., 2011; Wilson et al., 2007). In a North American species, *Faxonius palmeri longimanus*, the strength of a claw varies greatly among individuals of the same size (Robinson & Gifford, 2019), but no one has confirmed whether these claws function as signals. In two other species (*F. virilis* and *Procambarus clarkii*), competitors ignore claw size when escalating aggression (Graham & Angilletta, 2020; Graham et al., 2020), making large claws an unreliable indicator of strength but not a dishonest signal. This diversity of patterns underscores the need for comparative studies of aggressive behaviour to ascertain whether dishonest signalling occurs widely among species, or whether unreliability stems from noise inherent to a system (Carazo & Font, 2013; Wagner, 1992).

Here, we combined comparative and experimental methods to distinguish between unreliability and dishonesty in six species of freshwater crayfish. In each species, we studied the variation in claw size, claw strength and relative claw strength (residual of strength regressed onto size) between species that occupy distinct ecological niches: stream-dwelling species that use claws for fighting versus burrowing species that use claws for digging. Burrowing crayfish make an excellent comparative control, because they do not defend territories or fight with their claws (Dalosto et al., 2013; Guiaşu et al., 2005); therefore, the size and strength of a claw plays no role in signalling aggression, either honestly or dishonestly. On the other hand, stream-dwelling species fight frequently over resources, using their claws to grapple and pinch. Considering this difference in behaviour, we expected claw size and claw strength to vary more among individuals in species of stream-dwelling crayfish than in

species of burrowing crayfish. However, if relative claw strength and claw strength vary to similar degrees in stream-dwelling crayfish and in burrowing crayfish, we can conclude that processes other than selection for dishonest signalling produced the weak relationship between the size and strength of claws. Furthermore, because males fight more often and more intensely than females do (Moore, 2007), we expected greater variation in size and strength among males of stream-dwelling species than among females. Then, to establish whether claws of stream-dwelling species function as dishonest signals, we staged contests between males in a controlled environment. If large claws deter competitors from escalating aggression, we can conclude that claw size (or a close correlate) serves as a signal of fighting ability. In this way, the combination of experimental and comparative analyses enabled us to distinguish between unreliability and dishonesty in a system of animal communication.

2 | MATERIALS AND METHODS

2.1 | Study species

We studied six species of crayfish: three species described as primary burrowers (*Cambarus monongalensis*, *Cambarus dubius* and *Lacunicambus thomai*) and three species described as tertiary burrowers (*Cambarus carinirostris*, *Cambarus robustus* and *Faxonius obscurus*). The docile primary burrowers (hereafter referred to as burrowing crayfish) construct extensive burrows in which they spend most of their life (Hobbs, 1981). The aggressive tertiary burrowers (hereafter referred to as stream-dwelling crayfish) construct rudimentary burrows under refuge (Hobbs, 1981). Males and females of all six species possess enlarged frontal claws that perform distinct functions. Burrowing species do not defend or fight over burrows and lack ritualized behaviours associated with aggression (Dalosto et al., 2013; Guiaşu et al., 2005). By contrast, stream-dwelling crayfish possess enlarged claws that are used as weapons and potentially as signals of fighting ability (Moore, 2007; Robinson & Gifford, 2019; Wilson et al., 2007). In some stream-dwelling species, males produce large but weak claws that dishonestly signal their strength to competitors (Robinson & Gifford, 2019; Wilson & Angilletta, 2015). Variation in the size or performance of claws within burrowing species serves as a reference to the degree of variation expected in stream-dwelling crayfish; because the claws of burrowing species are not adapted for territorial signalling. Therefore, these two groups of species enable one to contrast the variation in a trait that has been selected as a signal and a trait that plays no role in signalling.

Although burrowing species are not thought to possess ritualized aggressive behaviour, aggression between stream-dwelling species is described based on the escalation from low-intensity to high-intensity behaviours (Moore, 2007). Initially, crayfish face each other with their claws opened and use antennae to tap each other's carapace. If these behaviours do not cause one crayfish to flee, the pair will approach each other and interlock claws, pushing periodically. If aggression escalates, the crayfish will pinch with their claws

until one of the competitors retreats with a tail flip. The highly ritualized fighting behaviour of stream-dwelling species makes them ideal candidates for studies of aggression and signalling within the laboratory environments. Furthermore, crayfish aggression in the laboratory is similar to aggression within natural settings (Bergman & Moore, 2003) and competitors will engage in competition during both the day and the night.

2.2 | Collection and husbandry of animals

We collected 359 crayfish representing the six species throughout Pennsylvania and West Virginia, USA from June 2019 to August 2019 (Table S1). All stream-dwelling species were either captured by hand or by net. By contrast, burrowing species construct burrows in substrate close to the water-table. Thus, burrowing species were caught by flooding burrows with water and plunging the burrow with a closed fist. Usually, the resident crayfish surfaced to investigate the disturbance, enabling one to capture the animal. Often, multiple bouts of plunging were required to capture a burrowing crayfish. Because burrowing species are harder to locate and capture, we collected fewer individuals of burrowing species than we did for stream-dwelling species (see Table S1). For this study, we only collected adult crayfish during their inter-moult period, such that all exoskeletons were hardened. Thus, we excluded any crayfish that showed signs of recent or current moulting, such as a soft exoskeleton or cracks in the carapace. The sex of each crayfish was identified by the presence or absence of male gonopods.

Each crayfish was held in a plastic container (14 × 9 × 6 cm) with 5 cm of dechlorinated tap water. Prior to any data collection, all crayfish were acclimated to laboratory conditions for 2 days to minimize any effect of nutritional state or social experience (Bergman et al., 2003; Daws et al., 2002; Graham et al., 2020). While housed in the laboratory, crayfish were not fed and were kept on a 14:10-hr light:day cycle. After data collection, each crayfish was returned to the location of capture.

2.3 | Variation in the size and strength of claws

The claws of each crayfish were photographed on the background of white graph paper (mm²). Images were imported to ImageJ (Schneider et al., 2012), a software that enabled us to quantify six linear dimensions of each claw: (a) width at heel, (b) width at dactyl/manus joint, (c) length of manus from heel to joint, (d) width of pollex at dactyl joint, (e) width of dactyl and (f) length of pollex (Figure S1). We converted the six dimensions to principal components, yielding a first principal component (PC1) describing more than 93.92% of the variation (Table S2). As in previous studies of crayfish (Bywater et al., 2008; Graham et al., 2020; Wilson et al., 2007, 2009), the scores of the PC1 were used as a measurement of claw size. To relate claw size to body size, we estimated body size as the length of the cephalothorax measured with digital calipers.

We measured the strength of each claw with a custom-built force transducer, consisting of two metal plates connected to a load cell and an amplifier. When held close to the plates, a crayfish used its claw to pinch the plates together. The amplifier generated a voltage that was transformed linearly to determine pinching force. We estimated maximal strength of each claw from the best of five consecutive pinches. Although studies have highlighted the pitfalls of measuring maximal performance (Losos et al., 2002), such measurements have been highly repeatable for other species of crayfish (Graham et al., 2020; Wilson et al., 2009). To confirm this repeatability in our study, we remeasured maximal claw strength on 2 consecutive days for a subset of individuals of each species.

2.4 | Analyses of claw strength

When analysing the repeatability of claw strength, we conducted a GLMM to determine the relationship between maximal strength recordings across consecutive days. Because we had data for both claws of each crayfish, we included the identity of crayfish as a random factor in the model. We also computed the intra-class correlation between measurements using the *irr* library in R (Gamer, 2010).

To quantify the residual variation in claw strength for a given claw size, we regressed the maximal claw strength onto the PC1 of claw size (scores of PC1). As in previous studies, we compared the fit of models in which the error was either fixed, exponential, power or constant power (Graham & Angilletta, 2020; Graham et al., 2020). To do so, we used the *nml* library in R (Pinheiro et al., 2017) to fit linear models of claw size and claw strength. The most likely model based on the corrected Akaike information criterion (AICc) was used to calculate the residual strengths of each claw.

Having estimated residual claw strengths, we then fit models with maximal claw strength as the dependent variable, and with an interaction between claw size, lifestyle and sex as independent variables. Furthermore, we included a random intercept associated with individual crayfish. To test the hypothesis that the burrowing lifestyle would lead to reduced selection for claws with immense variation in strength, we compared models with and without lifestyle as a fixed factor. When testing our hypothesis about how sex, species and lifestyle influence the variation in crayfish claw strength for a given size, we estimated the variance of claw strength for each combination of sex and lifestyle, to see whether adding these parameters provided a better fit to the data than a single variance for all groups. Based on our hypothesis, we expected that the most likely model would be one with a larger variance of claw strength for stream-dwelling crayfish than for burrowing crayfish. We also predicted a greater variance in claw strength for males than for females.

2.5 | Social dominance in staged encounters

A subset of the males of each stream-dwelling species was used to stage encounters between competitors. Burrowing crayfish were not

used during these trials because they neither behave aggressively nor defend territories (Dalosto et al., 2013; Guiaşu et al., 2005). Prior to staged encounters, we randomly selected half of the crayfish to be focal individuals and the other half to be opponents. Each focal individual experienced two to four staged encounters. Opponents were assigned randomly to fight with a focal crayfish with the constraint that no opponent could encounter the same focal crayfish more than once. At least 24 hr elapsed between the consecutive encounters for each focal individual. This procedure enabled us to collect a substantial amount of data from fights, while also controlling for any effects of prior experience (i.e. a winner or loser effect; see below).

In total, we staged 150 encounters: 40 encounters between males of *C. robustus*, 50 encounters between males of *C. carinirostris* and 60 encounters between males of *F. obscurus*. Staged encounters were conducted using methods from previous studies (Graham & Angilletta, 2020; Graham et al., 2020). Encounters occurred in a glass aquarium (35 × 20 × 15 cm) filled with dechlorinated water and divided by a removable screen. The screen prevented physical contact between crayfish without preventing visual or chemical signalling. We placed a pair of crayfish in the aquarium simultaneously and removed the divider after 5 min. Prior to removing the divider, crayfish rarely moved throughout the arena and never displayed claws. Upon removing the divider, crayfish typically engaged in aggression within 5 min. Aggression typically consisted of multiple bouts of interlocking claws and grappling, with occasional pinching. Once aggression ensued, we observed the interaction until the winner chased and the loser fled (Wilson et al., 2007). Upon determining a winner, we stopped the trial to prevent injuries. Importantly, some contests were settled solely by displays of claws. Thus, we categorized each encounter according to its final stage of escalation: (a) encounters that ended without physical contact, and (b) encounters that ended after physical contact (e.g. tapping, grappling or pinching with claws). For our analyses, we considered any encounter that escalated to physical contact as a fight. Collecting data on the final stage of aggression enabled us to disentangle factors promoting dominance prior to fighting and factors promoting dominance during fighting.

When testing our hypothesis about the traits that determine the outcome of territorial aggression in the stream-dwelling species, we modelled the factors influencing the probability of the focal crayfish engaging in aggression and the probability of winning a fight. We fit GLMMs to our data using the lme4 library of the R Statistical Package (Bates, 2010). In each case, the dependent variable was coded as a discrete outcome (0 or 1); therefore, we used a model with a binomial distribution of error. To control for the effects of collinearity in our fixed effects, we initially ran separate models of carapace size, claw size and claw strength difference to see how these effects influenced out independent variables (probability of fighting or probability of winning). Because each focal crayfish engaged in more than one encounter with unique opponents, we used focal crayfish as a random factor in our analysis. Including the identity of the focal crayfish in our models enabled us to control for variation in behaviour

among focal crayfish and account for potential winner effects or loser effects (Graham & Angilletta, 2020; Graham et al., 2020). Additionally, a random slope of species was added to these models. The single term model with the lowest AIC value was then used for consequent analyses, described below. After determining the best model from the models comparing our fixed effects, we used multimodel averaging to estimate the most likely value of means. Then, we used the MuMIn library (Bartón, 2018) to fit all possible models to the data. After fitting all possible models, we calculated the AIC and Akaike weight of each model, the latter variable being the probability that the model best describes the data. Finally, we calculated the weighted average of each parameter including estimates from all models. The resulting values of parameters were used to calculate the most likely mean for each combination of factors. This approach eliminates the need to interpret *p*-values because all models (including the null model) contributed to the most likely value of each mean.

3 | RESULTS

Both burrowing species and stream-dwelling species had a substantial variation in claw strength, but the majority of this variation was unrelated to claw size (Table 1; Figure 2). Two factors rule out measurement error as a primary cause of residual variation in claw strength. First, claw strength was highly repeatable between consecutive days (Figure 3; Table S3). For five of the six species, the intra-class correlation of consecutive measurements ranged from 0.93 to 0.97; for the sixth species, *Cambarus monongalensis*, the correlation was 0.84. These high values of repeatability reflect the precision of our strength measurements. Second, the principal component score of claw size (PC1) correlated strongly with body length (Table S4), suggesting that our morphological measurements were

TABLE 1 Results of covariation between claw size and claw strength for the six species included in our study. Across all species, the force measurement that we recorded exhibited substantial variation in strength for a given size; depicting by the low values of r^2 for each species. M, male; F, female; SD, stream-dweller; B, burrower

Species	Lifestyle	Sex	r^2	<i>p</i>
<i>Cambarus carinirostris</i>	SD	M	0.26	<0.001
		F	0.13	<0.001
<i>Cambarus robustus</i>	SD	M	0.23	<0.001
		F	0.15	<0.002
<i>Faxonius obscurus</i>	SD	M	0.36	<0.001
		F	0.30	<0.001
<i>Cambarus monongalensis</i>	B	M	0.06	0.209
		F	0.26	<0.001
<i>Cambarus dubius</i>	B	M	0.001	0.869
		F	0.27	<0.001
<i>Lacunicambus thomai</i>	B	M	0.08	0.273
		F	0.19	0.014

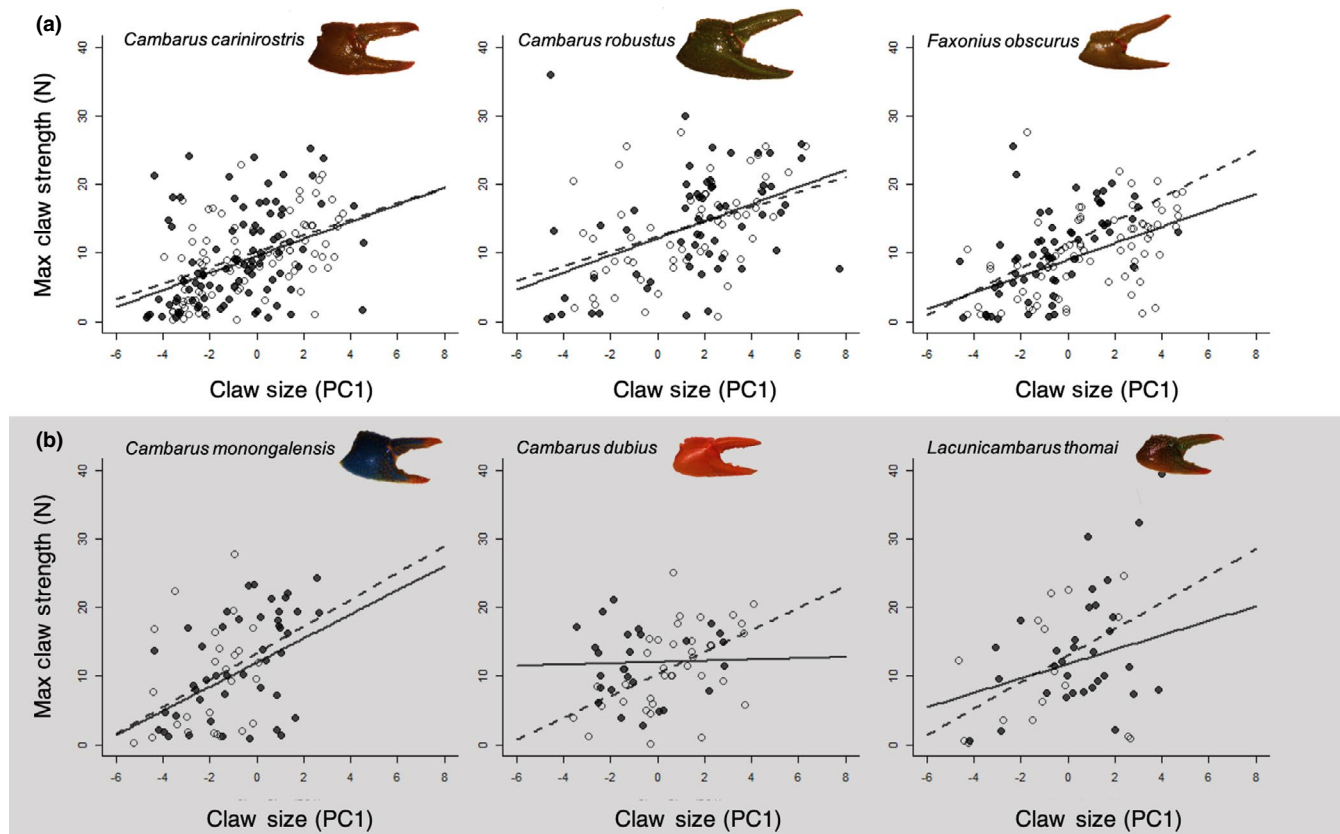


FIGURE 2 Relation between claw size (PC1) and maximal claw strength in three (a) stream-dwelling and the (b) three burrowing species included in our study. Filled and unfilled circles represent data for males and females respectively. Furthermore, the full black line and dotted black line represent the linear relationship between males and females respectively. Species within the grey shaded area are the burrowing species, whereas the non-shaded areas represent the stream-dwelling species

also accurate. Surprisingly, burrowing crayfish had even greater residuals of claw strength than did stream-dwelling crayfish (Figure 4). This difference was reflected in the standard deviations of strength in the linear model: 6.8 N and 6.6 N for males and females of burrowing crayfish, respectively, compared to 5.0 N and 6.1 N for males and females of stream-dwelling crayfish.

Regarding the territorial contests in the stream-dwelling species, the difference in body length predicted the winner better than the difference in claw size did in fights that did not escalate to physical contact (Figure 5a; Table S5). Likewise, the same pattern was observed when we analysed the outcome of contests that did escalate to fighting; the crayfish with a larger body length was more likely to win a fight (Figure 5b; Table S7). The identity of the species had a small effect on the outcome of a contest, suggesting only subtle variation in aggressive behaviour among species (Figure 5; Tables S5 and S7).

4 | DISCUSSION

Contrary to our prediction, we found that both stream-dwelling species and burrowing species have a substantial variation in claw size and residual variation in claw strength. In fact, the burrowing species

have even greater variation in claw strength compared to stream-dwelling species. Importantly, the claw strength measurements that we recorded were highly repeatable between days, which suggests that measurements of maximal pinching strength reflected pinching ability more than pinching motivation. For these reasons, we believe that some biological process, as opposed to measurement error, underlies the extensive residual variation in claw strength that we observed.

Researchers often interpret residual variation in an index of quality as evidence of dishonesty (Caro et al., 2016; McGuire et al., 2018), but this interpretation assumes that such variation stemmed from selection for signals that benefit the signaller at the expense of the receiver. The greater residual variation in burrowing crayfish cannot be interpreted as evidence of dishonest signalling, because these crayfish do not engage in territorial or aggressive signalling (Dalosto et al., 2013; Guiaşu et al., 2005). Unlike the aggressive stream-dwelling species, burrowing crayfish species completely ignore one another when paired (lead author, pers. obs.). Furthermore, we found equal rates of unreliability between the claws of males and the claws of females, although males signal less honestly than females in other species of crayfish (Robinson & Gifford, 2019; Wilson et al., 2007). Therefore, a large degree of residual variation, on its own, constitutes poor evidence of dishonest signalling in a population. Perhaps more

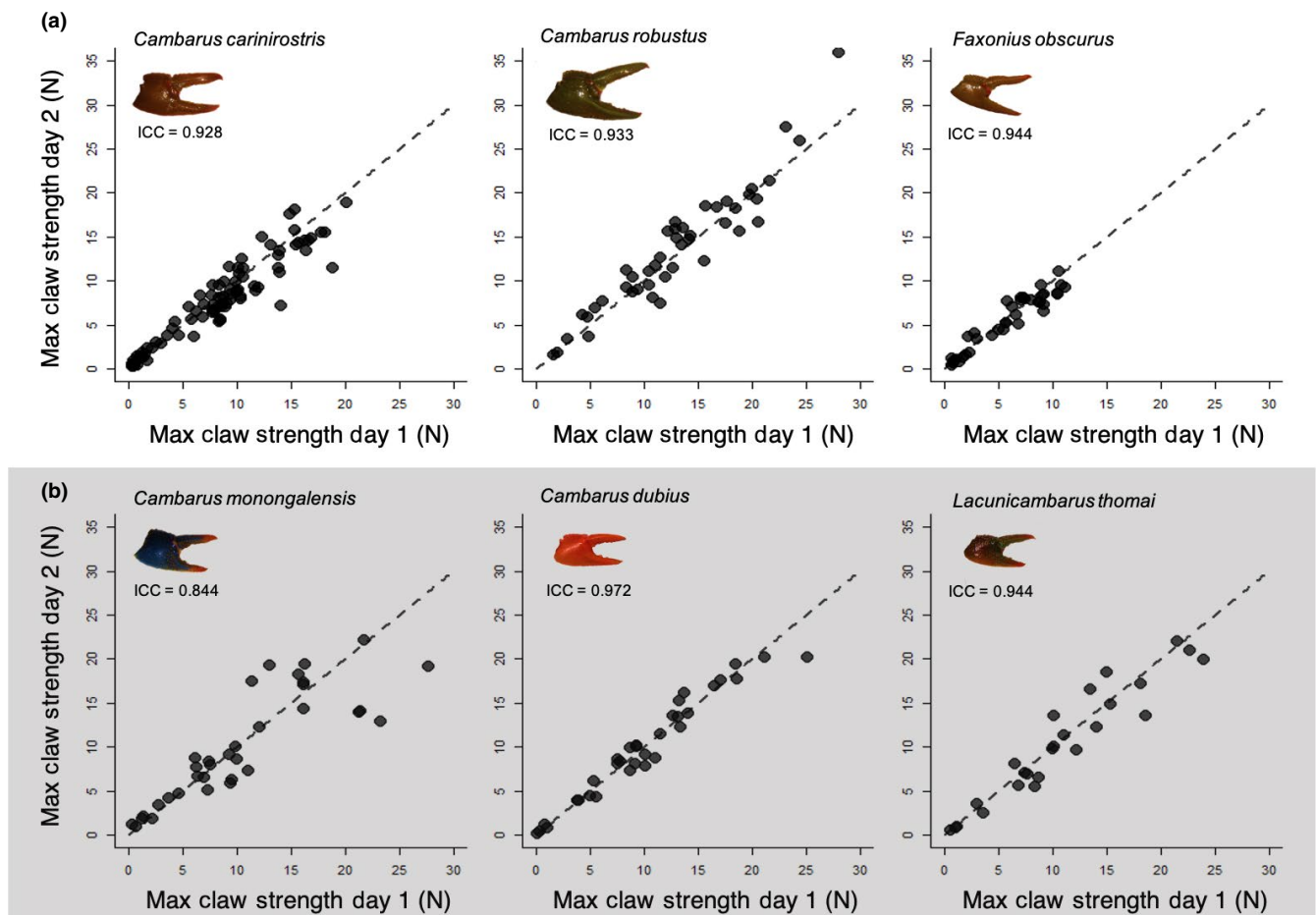


FIGURE 3 Relation between the strength of the claw in males and females of (a) the three stream-dwelling species and (b) the three burrowing species on day 1 and day 2 controlling for crayfish identity as a random factor in a GLMM. The dashed line serves as a reference for the equality of strength between days. Species within the grey shaded area are the burrowing species, whereas the non-shaded areas represent the stream-dwelling species

importantly, stream-dwelling crayfish did not escalate aggression according to claw size. In staged contests, the probability of fighting was unrelated to the difference in claw size between crayfish. These patterns differ from those observed in studies of *Cherax* species of crayfish, in which a male with larger claws was more likely to chase away a competitor or win a fight (Seebacher & Wilson, 2007; Wilson et al., 2007); in those species, the relationship between claw size and social dominance lent credence to the view that claw size served as a signal of competitive ability.

Several factors other than dishonest signalling could account for the substantial variation in claw strength of crayfish. First, the claws might have been selected to function purely as weapons, with little or no signalling functions (McCullough et al., 2016). In this way, selection may not favour individuals with large and proportionately strong claw. Instead, traits other than claw size and claw strength may better predict fighting success, such as the length or mass of the body (Graham et al., 2020; Ueno & Nagayama, 2012). Second, despite the different functions of claws in burrowing crayfish and claws of stream-dwelling crayfish, these groups share a developmental process that could create variation

in strength for a given size. Crayfish must moult their exoskeleton to grow, with some individuals moulting several times per year. The largest muscle of the claw undergoes substantial atrophy of muscle before moulting (West, 1997). Although the exoskeleton of crayfish hardens quickly after a moult, a crayfish needs a longer period to regenerate atrophied muscles. Thus, crayfish must periodically regenerate muscles and cuticles of the claw (Chang, 1995). Because the strength of crustacean claws is likely determined by the time since the last moult, a large difference in strength can occur between claws of similar size. In our study, we controlled for variation in strength due to moulting by only including individuals in a hardened inter-moult stage. Both burrowing crayfish and stream-dwelling crayfish moult regularly, potentially explaining the variation in claw strength that we observed. In species of crayfish that use claws as signals, such as *Cherax dispar*, moulting may generate the variation in claw strength that enables dishonest signalling. In other species, such as the ones that we studied, the variation in claw strength may be too great for dishonest signalling to evolve; a signal must have a minimum degree of reliability to offer value to receivers. These explanations demonstrate that

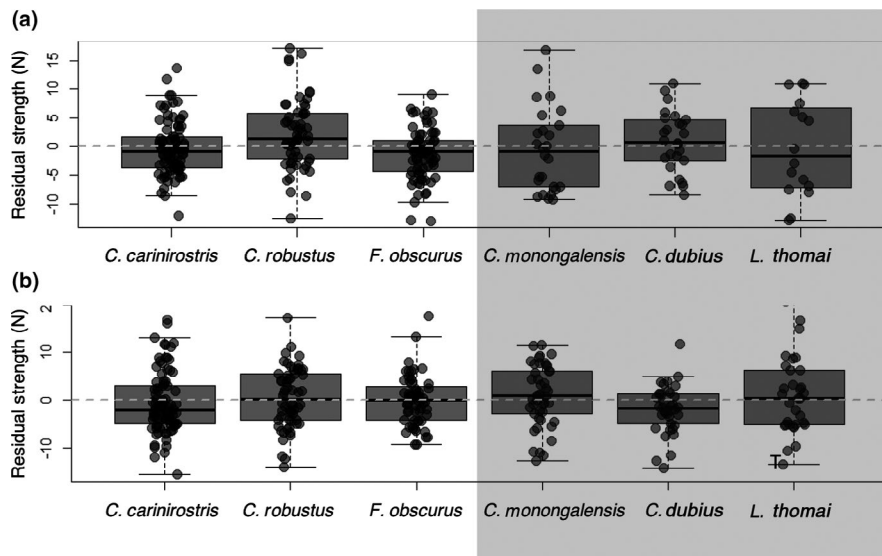


FIGURE 4 The residual maximal claw strength for (a) male and (b) females of the six species included in our study. Species within the grey shaded area are the burrowing species, whereas the non-shaded areas represent the stream-dwelling species. Contrary to our prediction, the residual variation in claw strength was greater in burrowing species. Error bars represent 95% confidence intervals around the mean residual strength for each species. The dashed references line depicts a residual strength value of zero

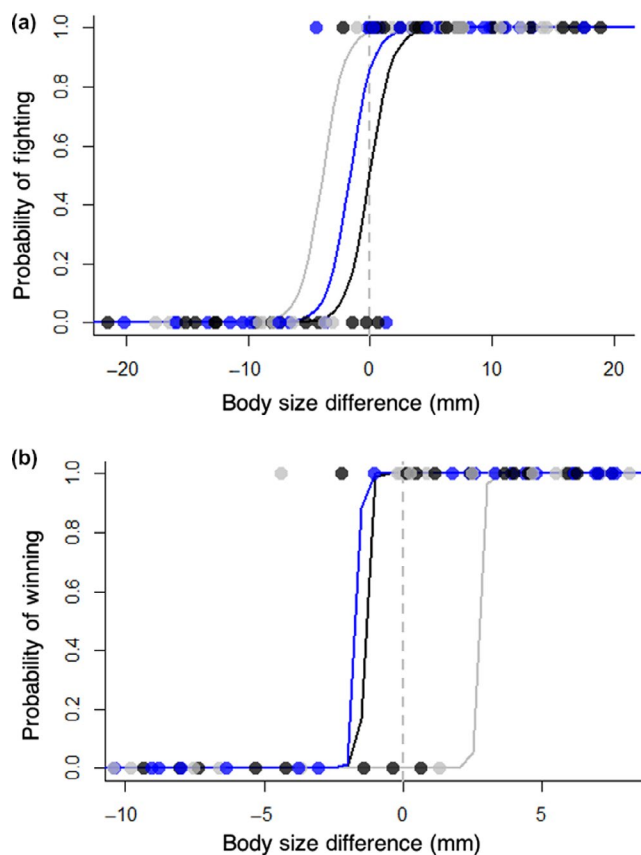


FIGURE 5 (a) Probability of escalating aggression to physical combat based on the difference in body size between opponents. (b) Probability of winning a fight based on the difference in body size between opponents. Circles indicate the outcome of each encounter (0 or 1, where 1 represents winning). The lines represent the fitted values of statistical model that describes our data. In both graphs, the black dots and lines represent data from *Cambarus carinirostris*. Grey dots and lines represent data from *Cambarus robustus*, and blue dots and lines represent data from *Faxonius obscurus* aggression. The dotted line provides a reference for zero difference in body size

processes other than dishonest signalling can generate residual variation in claw strength. Similar developmental or evolutionary processes must be considered when determining the honesty of animal signals.

Our results call for biologists to distinguish unreliable indicators from dishonest signals when studying animal communication (Wilson & Angilletta, 2015). Dishonest signals have evolved to convey unreliable information by weakening the correlation between the signal and the information conveyed (Searcy & Nowicki, 2005). By contrast, unreliable indicators may arise from processes other than sexual selection of signals, such as development noise or genetic drift (Botero et al., 2010; Brumm, 2013; Tanner & Bee, 2019). In other words, all dishonest signals are unreliable indicators, but not all unreliable indicators are dishonest signals. Although some crustaceans seem to signal strength dishonestly (Bywater et al., 2015; Bywater & Wilson, 2012; Wilson & Angilletta, 2015), we now have evidence suggesting that the claw size of a crayfish can unreliably indicate their competitive ability, but is unlikely to serve as a signal during aggression. This finding should prompt other researchers to re-evaluate assumptions about the role of putative dishonest signals in communication by other species.

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CONFLICT OF INTEREST

The authors declared no conflict of interest.

AUTHORS' CONTRIBUTIONS

This study was conceived by Z.A.G. and M.J.A.; Z.A.G. collected the data; Z.A.G. and M.J.A. analysed the data, drafted and edited the manuscript.

DATA AVAILABILITY STATEMENT

Data available from the Dryad Digital Repository <https://doi.org/10.5061/dryad.18931zcz7>. R scripts will be made available upon request.

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